Substrate Stability Under Persistent Disagreement: Structural Constraints for Neutral Ontological Substrates (in-progress)

Denise M. Case

Northwest Missouri State University, Computer Science and Information Systems, Maryville, MO, USA

March 12, 2026

Abstract

Modern data systems increasingly operate under conditions of persistent legal, political, and analytic disagreement. Under such conditions, interoperability cannot rely on shared interpretation, negotiated semantics, or centralized authority. Instead, representations must function as neutral substrates capable of preserving stable reference across incompatible extensions.

This paper analyzes the structural constraints that the requirement of stable reference under persistent disagreement imposes on ontological design. Treating interpretive neutrality and extension-stable identity as explicit design constraints, we ask what minimal ontological structure is forced if accountability relationships must remain referable and comparable under disagreement. Minimality is defined structurally: reductions are admissible only if they do not reintroduce identity-relevant distinctions through hidden roles, flags, or contextual predicates.

The paper establishes a conditional lower bound: any ontology capable of preserving stable reference under persistent disagreement

must realize at least six distinct identity-and-persistence regimes. A construction with exactly six regimes is shown to satisfy the stated requirements without embedding causal or normative commitments in the substrate. The result is not a proposal for a universal ontology but a constraint on ontological design under persistent disagreement.

Statement of Result (Preview). *Under explicit neutrality and stability constraints, attempts to simplify ontological structure by collapsing distinctions reintroduce those distinctions as hidden interpretive mechanisms. Excluding such hidden regimes implies six distinct identity-and-persistence regimes for accountability under persistent disagreement: fewer do not satisfy the stated requirements, and additional regimes introduce no further structural capacity.*

Informally: Neutrality is achieved by externalizing explanatory and evaluative structure to extension layers. To function as a substrate for disagreement, an ontology must remain agnostic with respect to causal and normative interpretation.

The formal necessity argument appears in Section 4, and the sufficiency construction in Section 5.

Keywords: formal ontology; ontological neutrality; accountability; neutral substrates; extension stability; causal and normative commitment

1 Introduction: Structure Under Disagreement

Modern data systems must often support stable reference and accountability across settings characterized by persistent legal, political, and analytic disagreement.Three recurring settings make the structural requirements concrete. First, in multi-jurisdiction regulatory reporting, a single corporate entity operates under overlapping legal regimes that individuate obligations,

violations, and responsible parties differently; audit trails must remain comparable across jurisdictions without presupposing which regime governs. Second, in legislative history tracking, statutory authority is amended, delegated, and contested over time, requiring stable reference to normative structures independently of the occurrences that enact or challenge them. Third, in cross-institutional provenance analysis, research or administrative records are produced, transferred, and re-analyzed across institutions whose classificatory and causal assumptions diverge, yet must remain referable as the same records across those divergent frameworks. These cases are not offered as empirical evidence; they function as structural diagnostics that make concrete the requirements of Section 3.

Upper ontologies and general-purpose representational frameworks including BFO (Arp et al. 2015a), DOLCE (Gangemi et al. 2002), and PROV-O (Lebo et al. 2013) are expressive and well-motivated for their intended purposes. They support rich domain modeling, causal attribution, and normative classification. However, expressiveness is not neutrality. These frameworks were designed to represent the world accurately within an interpretive context, not to remain agnostic across incompatible interpretive contexts simultaneously. When deployed as shared substrates under persistent disagreement without additional constraints, their embedded commitments such as what counts as an agent, what individuates an event, and what constitutes an authority, can become sites of interpretive conflict rather than stable reference points.

The problem addressed here is distinct from the general ontology design problem. The objective is not representational adequacy within a given framework but substrate stability across frameworks that cannot be assumed to converge. No single interpretive authority can be presupposed, and semantic convergence cannot be enforced without excluding legitimate participants or perspectives. Nevertheless, shared use of data remains necessary: entities must be referable, actions traceable, and records comparable across frameworks that may diverge in assumptions, methods, or evaluative standards. Recent work confirms that systems without explicit structural commitments produce unstable and context-dependent categorical behavior (Köhler and

Neuhaus 2025), further motivating the need for substrates that enforce stable reference structurally rather than relying on interpretive consistency.

A natural response might be to reduce ontological commitment by collapsing distinctions and representing entities, occurrences, authorities, and records as a single undifferentiated type, differentiated only by role predicates (e.g., *isAgent*, *isOccurrence*, *isAuthority*) or using contextual flags supplied at query or extension time. This achieves surface parsimony but not structural neutrality. Such encodings relocate essential distinctions from the substrate to interpretive mechanisms that are not represented at the substrate level, and whose consistent application requires precisely the shared commitments that persistent disagreement denies. We refer to these as *hidden regimes*, structural discriminations that are operationally necessary but ontologically unacknowledged. They are incompatible with the design objective, because they make referential stability contingent on interpretive consensus that cannot be assumed.

The preceding analysis identifies a design objective that is distinct from the general problem of ontology construction. We adopt a neutrality framework developed in prior work (Case 2026), treating ontological neutrality as a structural design constraint. Neutrality is understood as interpretive non-commitment at the substrate level combined with stability under incompatible extension: a substrate must permit independent extensions to introduce causal, normative, or analytic structure without requiring revision of the underlying representation. Explanatory and evaluative structure is externalized; the substrate is constrained to remain pre-causal and pre-normative.

The substrate is not intended to adjudicate causal or normative claims. Its role is limited to preserving referential comparability across independently evolving extensions. All causal explanation, normative evaluation, and domain-specific interpretation are delegated to extension layers.

Stable reference in this work concerns regime-level persistence conditions, not immunity to transformation or versioning at interpretive layers. The

substrate must preserve the referential integrity of entities and records even as their interpretation evolves in incompatible ways across extensions. Persistent disagreement, as used in this work, is defined by four conditions:

- Extensions may introduce incompatible causal accounts.
- Extensions may introduce incompatible normative attributions.
- Extensions cannot be forced into convergence.
- Substrate revision is not permitted to resolve interpretive conflict.

Under these conditions, the substrate must remain interpretively agnostic while preserving referential stability across extensions. Stability is a property of the identity-and-persistence regime, rather than any particular token or interpretive assignment within it.

Within this setting, minimality is not a reduction in the number of primitives. Apparent simplifications may achieve surface parsimony by reintroducing essential distinctions indirectly through roles, flags, contextual typing, or other internal discriminators interpreted at extension layers. Such encodings do not eliminate structure; they relocate it to mechanisms that presuppose shared interpretive commitments. Under persistent disagreement, those commitments cannot be assumed. Any reduction that preserves stability only by deferring identity-and-persistence distinctions to extension layers therefore fails the design objective. Minimality is thus defined as the exclusion of hidden coordination mechanisms: all operationally necessary identity-and-persistence distinctions must be explicit and structurally enforced at the substrate level.

Accountability is the specific capacity discussed in this paper. Accountability analysis requires that a neutral substrate support stable reference to participants, authority structures, regulated occurrences, applicability contexts, and descriptive records, without embedding causal claims or normative judgments in the substrate itself. These are not introduced as a proposed ontological partition, but as representational capacities whose simultaneous support places structural constraints on any neutral substrate. Whether they

can be realized within fewer identity-and-persistence regimes is addressed in Section 4; the question is whether a neutral substrate can support them at all, and if so, with what irreducible structural commitments. Three diagnostic cases (multi-jurisdiction regulatory reporting, legislative history tracking, and cross-institutional provenance) recur throughout the paper to make these requirements concrete; they are structural diagnostics, not empirical evidence.

The central contribution of this paper is a conditional bound on ontological structure given the design objectives outlined above. Under these neutrality and stability constraints, we show that any ontology capable of supporting accountability under persistent disagreement must realize at least six distinct identity-and-persistence regimes. We further show that a construction realizing exactly six such regimes is sufficient to satisfy the stated requirements without introducing hidden or implicit structure. The result is not a proposal for a universal ontology but a constraint on what is structurally possible under a specific optimization objective.

Alternative objectives prioritize different tradeoffs, such as interpretive convergence, domain-specific efficiency, or context-dependent classification at the substrate level; these are not discussed here.

The contribution of this paper is a constraint result. The question addressed is structural rather than taxonomic: if a representational substrate must support accountability across independently evolving interpretations while preserving stable reference to shared entities and records, what identity-and-persistence structure, if any, is unavoidable?

Treating neutrality and extension-stable reference as explicit design constraints and grounding the analysis in concrete accountability workflows (regulatory reporting, legislative history tracking, and cross-institutional provenance), the paper derives a conditional lower bound on the number of regimes required.

The remainder of the paper proceeds as follows. Section 2 states the assump-

tions, neutrality constraints, and optimization criterion. Section 3 formalizes the structural requirements imposed by accountability analysis. Section 4 derives the lower bound. Section 5 presents a construction achieving this bound. Section 6 discusses the scope and implications. Section 7 situates the work relative to existing frameworks. Section 8 concludes.

2 Assumptions, Neutrality, and Optimization Criterion

This section fixes the problem setting for the remainder of the paper. All results that follow are conditional on the assumptions stated here. The purpose is not to argue that these assumptions are universally correct, but to make explicit the constraints under which a neutral substrate for accountability is possible.

2.1 Persistent Disagreement and Uncoordinated Extension

We assume environments in which disagreement is not an anomaly to be resolved, but a persistent condition. Legal interpretation, policy evaluation, and analytic methodology may remain incompatible across communities and over time. Accordingly, no single interpretive authority or shared semantic framework is assumed at the substrate level.

We further assume that extensions to the substrate may occur without coordination. Independent actors may add explanatory, normative, or analytic structure, and such extensions must not require revision of the substrate itself. This assumption does not rely on global semantic agreement, centralized governance, or negotiated meaning as preconditions for interoperability. Under these assumptions, substrate-level admissibility must not depend on interpretive agreement among extensions.

2.2 Neutrality Constraint

This work assumes substrate neutrality and stability as a design constraint rather than as a philosophical claim. A neutral substrate does not assert why events occur, whether outcomes are desirable, or how actions ought to be evaluated. Instead, it provides stable referents and structural relations upon which external interpretive layers may operate.

Neutrality further requires stability under incompatible extension at the level of identity conditions and admissible relations. If two extensions disagree in interpretation, evaluation, or explanation, their disagreement must not force reclassification of entities, alteration of identity conditions, or revision of the substrate. Any representation whose admissibility conditions depend on interpretive agreement does not satisfy this constraint.

2.3 Stable Reference Requirement

For a substrate to support accountability, entities must remain referable over time and across extensions. Identity conditions and admissible relationships must not depend on interpretive convergence under changes in interpretation. If an entity must change its ontological status in order to participate in new relationships or analyses, stable reference is not preserved. This requirement is not satisfied by representations in which identity is determined contextually or retroactively by interpretive layers. If identity conditions shift when an institution introduces a new classificatory scheme, the provenance chain breaks and accountability for the record's content and custody cannot be maintained.

This requirement excludes representations in which stability-critical distinctions are encoded only as roles, flags, or contextual predicates. Such mechanisms rely on interpretive conventions supplied by extension layers and therefore permit divergent admissibility judgments under persistent disagreement.

The stability requirement is intentionally strong. Many engineering systems address identity pressure through contextual or negotiated identity schemes. The present analysis instead examines the case where such mechanisms are unavailable and stable reference must be preserved independently of interpretive convergence.

2.4 Optimization Criterion

The optimization objective of this paper is to minimize explicit ontological structure subject to the constraints above. Minimality is not defined as the smallest possible number of primitives in the abstract, but as the smallest number of explicit identity-and-persistence regimes required to preserve neutrality and stable reference.

A proposed reduction is counted as a simplification only if it does not reintroduce the same distinctions through hidden or interpretive mechanisms. Collapsing regimes while reintroducing their distinctions via internal discriminators, contextual typing, or role predicates does not reduce total structural complexity and is therefore excluded by the optimization criterion adopted here.

The minimality criterion adopted here is analytical rather than prescriptive: it is introduced to determine what ontological structure is unavoidable when neutrality and stability under persistent disagreement are treated as design constraints.

Alternative optimization objectives prioritize different tradeoffs, such as interpretive coordination, domain-specific efficiency, or context-dependent classification at the substrate level. The results presented here apply only to substrates optimized for interoperability under persistent disagreement.

3 Structural Requirements for Accountability Substrates

Given the assumptions and optimization criterion fixed in Section 2, the central question is not which ontology is most expressive or convenient, but which structural distinctions are unavoidable. We therefore state the representational requirements that an ontological substrate must satisfy in order to support accountability analysis under persistent disagreement.

These requirements are structural rather than interpretive. They do not prescribe how accountability should be understood, evaluated, or enforced; instead, they specify what must be representable in a stable and neutral manner.

The requirements are framed as necessary capacities. They are not derived from any particular domain, policy area, or theory of governance, but from the minimal conditions under which accountability claims can be stated, inspected, and compared without embedding causal or normative commitments in the substrate.

Before deriving the lower bound, we make explicit what is meant by an *identity-and-persistence regime*.

Definition 3.1. *Identity-and-Persistence Regime An identity-and-persistence regime specifies the conditions under which entities remain the same entities over time and across incompatible extensions, together with the relations they may admit without reclassification.

Two entities belong to the same regime if and only if they share the same stability-critical identity criteria, persistence conditions, and admissible relations under the neutrality and stability constraints adopted here.

Regimes are not introduced as ontological kinds or domain categories. They are structural distinctions forced by the requirement that identity and admissibility remain invariant under persistent disagreement.

3.1 R1: Stable Identity and Persistence

Requirement. The substrate must support stable reference to entities over time.

Each entity must have identity and persistence conditions that are invariant under incompatible extensions, independent of how the entity is later classified, interpreted, or evaluated. Changes in interpretation, role, status, or evaluation must not require reclassification of entities or modification of their identity criteria. This requirement does not apply to representations in which identity is determined contextually or retroactively by interpretive layers. If an entity must change its ontological status in order to participate in new relationships or analyses, stable reference is not preserved.

In cross-institutional provenance, a research record transferred between institutions must remain referable as the same record regardless of how each institution classifies, annotates, or re-analyzes it. If identity conditions shift when an institution introduces a new classificatory scheme, the provenance chain breaks and accountability for the record's content and custody cannot be maintained. R1 is therefore non-optional wherever records must persist as stable referents across frameworks that may individuate them differently.

3.2 R2: Obligation-Bearing Capacity

Requirement. The substrate must support representation of entities that can bear obligations or responsibilities independently of particular occurrences.

Such entities must be referable independently of the particular occurrences in which obligations are fulfilled, violated, or discharged. This requirement does not presuppose any substantive theory of normativity. It requires only that obligation-bearing entities be representable as enduring referents whose identity does not depend on specific events.

In multi-jurisdiction regulatory reporting, a corporate entity must be referable as an obligation-bearer across legal regimes that may differ in how they define corporate personhood, subsidiary responsibility, or delegated liability. If the substrate cannot represent obligation-bearing entities independently of the specific events in which obligations are fulfilled or violated, cross-jurisdiction audit comparisons require interpretive agreement on the nature of the bearer itself, precisely the agreement that cannot be assumed under persistent disagreement.

3.3 R3: Normative Reference Without Execution

Requirement. The substrate must distinguish between the existence of authority or obligation and the occurrences that execute, comply with, or violate it.

Normative or regulatory structures must therefore be representable independently of the occurrences that enact, comply with, or violate them. This distinction is necessary to support accountability over time, including cases of delayed enforcement, partial compliance, amendment, or contested application. Encoding normative reference solely through occurrences collapses authority into execution and undermines stable reference to governing structures.

In legislative history tracking, a statutory authority must be referable as a normative structure independently of the legislative acts that enacted it, the administrative actions that applied it, and the judicial decisions that interpreted it. If normative reference is encoded solely through occurrences, a statute that has been amended but not yet enforced, or contested but not yet resolved, has no stable substrate-level referent. Accountability for compliance with authority-as-written rather than authority-as-executed requires R3 as a structural precondition.

3.4 R4: Time-Indexed Occurrence

Requirement. The substrate must support representation of occurrences individuated by temporal realization and provenance.

Occurrences must be individuated by their temporal realization and provenance, and must be distinguishable from enduring entities. This requirement applies to actions as well as to other discrete occurrences relevant to accountability, such as filings, inspections, or transactions. Occurrences are not required to carry interpretive or evaluative meaning; they must only assert that something occurred at a particular time and place.

In multi-jurisdiction regulatory reporting, a filing, inspection, or transaction must be individuated by when and where it occurred, independently of how different jurisdictions classify or evaluate it. Two jurisdictions may disagree on whether a given action constitutes compliance, but both must be able to refer to the same occurrence as their object of evaluation. Without substrate-level individuation of time-indexed occurrences, that shared reference is unavailable and cross-jurisdiction comparison reduces to an interpretive dispute about what happened rather than about how to evaluate it.

3.5 R5: Applicability and Scope

Requirement. The substrate must support representation of applicability contexts that scope where, to whom, or under what conditions a normative structure applies.

It must be possible to state where, to whom, or under what conditions a normative or regulatory structure applies, without collapsing such contexts into either obligation-bearing entities or physical loci. Applicability contexts may be nested, overlapping, or partially coincident. They must therefore be representable as stable referents in their own right, rather than as incidental attributes whose interpretation may vary across extensions.

In multi-jurisdiction regulatory reporting, normative structures apply to different sets of entities depending on jurisdiction, and those jurisdictions may be nested, overlapping, or contested. If applicability contexts cannot be represented as stable referents in their own right, the scope of an obligation can only be recovered by interpreting the obligation itself which reintroduces the shared interpretive assumptions the substrate is designed to avoid. R5 is non-optional wherever accountability claims must be scoped without presupposing agreement on which regime governs.

3.6 R6: Descriptive Indicators Without Causal Commitment

Requirement. The substrate must support representation of descriptive indicators or records that characterize states, performance, or outcomes associated with entities or occurrences without asserting causal or normative claims.

Such indicators must be representable without asserting causal relations, evaluative judgments, or explanatory claims. This requirement is necessary to support longitudinal comparison and outcome inspection while preserving neutrality. Collapsing indicators into occurrences or into interpretive annotations eliminates stable reference to descriptive records over time and across analytic frameworks.

In cross-institutional provenance, performance indicators and outcome records associated with transferred data must remain comparable across institutions whose causal and evaluative frameworks differ. If descriptive indicators are encoded as occurrences, they carry implicit individuation conditions tied to causal assumptions; if encoded as interpretive annotations, they are unstable under extension. R6 is non-optional wherever longitudinal comparison of records must be preserved without committing the substrate to any particular explanatory or evaluative framework.

3.7 Summary

Together, R1–R6 specify the minimal representational capacities required for an accountability substrate operating under persistent disagreement. They do not yet impose a particular ontological partition. Rather, they define the constraints under which any such partition must operate.

In the following section, these requirements are used to derive a lower bound on the number of distinct identity-and-persistence regimes that a substrate must realize in order to satisfy them simultaneously.

4 Necessity: A Lower Bound on Ontological Structure

The previous section specified the structural requirements for accountability substrates under neutrality and extension-stable identity. We now use those requirements to derive a lower bound on the ontological structure that any such substrate must realize.

The result is a necessity claim: if a substrate is to support accountability under persistent disagreement while maintaining neutrality and stable reference, some minimum number of distinct identity-and-persistence regimes is unavoidable. The lower bound is established by showing that certain representational capacities cannot be realized within a single regime without violating extension-invariant identity or reintroducing hidden discriminators.

The regimes introduced below are persistence-and-admissibility profiles induced when the requirements of Section 3 must be satisfied simultaneously under the neutrality and stability constraints fixed in Section 2.

4.1 Goal and Proof Strategy

The goal of this section is to prove a numeric lower bound on the number of identity-and-persistence regimes required to satisfy R1–R6 simultaneously under the optimization criterion fixed in Section 2.

The argument has three steps.

First, we extract from R2–R6 (subject to R1) a set of structurally distinct persistence-and-admissibility profiles that must be realizable in any compliant substrate. These profiles correspond to candidate identity-and-persistence regimes forced by the requirements.

Second, we state a collapse admissibility criterion: a proposed identification of two regimes is admissible only if it preserves extension-invariant identity and admissible relations without introducing interpretive flags, roles, or context-dependent predicates at the extension layer.

Third, we show that any collapse among the forced profiles violates the admissibility criterion through one of three failure modes: identity instability, category error, or hidden-regime reintroduction. It follows that the substrate must realize at least as many distinct regimes as there are forced profiles, establishing the lower bound.

4.2 Induced Identity-and-Persistence Regimes

We now determine which identity-and-persistence regimes are forced by R2–R6 when satisfied simultaneously under R1's requirement of extension-invariant identity and admissible relations.

An identity-and-persistence regime is characterized by (i) persistence conditions and (ii) the class of relations admissible without reclassification.

When the requirements are satisfied simultaneously under R1, their persistence and admissibility constraints give rise to distinct candidate identity-

and-persistence profiles. We extract these profiles in turn.

4.2.1 Profile induced by R2: Obligation-bearing persistence

R2 requires representation of entities that bear obligations independently of particular occurrences. Such entities must persist across compliance, violation, delegation, and discharge. Their admissible relations include normative attribution without altering identity. This induces an initial distinct persistence-and-admissibility profile: obligation-bearing.

4.2.2 Profile induced by R3: Normative-structure persistence

R3 requires representation of normative or regulatory structures independently of their execution. Such entities persist across enactment, amendment, interpretation, and repeal. Their admissible relations include structural modification without collapsing into occurrences. This induces a second distinct profile: normative-structure.

4.2.3 Profile induced by R4: Time-indexed occurrence persistence

R4 requires representation of occurrences individuated by temporal realization and provenance. Such entities do not persist through modification but are fixed by their time-indexed realization. This induces a third distinct persistence profile incompatible with enduring entities: time-indexed occurrence.

4.2.4 Profile induced by R5: Applicability-context persistence

R5 requires representation of scopes or applicability contexts as enduring entities. Such entities admit nesting, overlap, and conditional scoping without collapsing into either authority or bearer entities. This induces a fourth

structurally distinct profile: applicability-context.

4.2.5 Profile induced by R6: Descriptive-record persistence

R6 requires representation of descriptive indicators without causal or normative commitment. Such entities admit measurement and comparison relations without implying execution or authority. This induces a fifth distinct profile: descriptive-record.

4.2.6 Enduring Non-Normative Referents

Satisfaction of R2–R5 presupposes stable reference to enduring entities that are neither obligation-bearing nor authority-bearing, yet may be regulated, inspected, or otherwise affected by occurrences. Such entities persist independently of normative attribution and independently of time-indexed realization. Their admissible relations differ from both normative-structure and obligation-bearing profiles. This yields a sixth distinct persistence-and-admissibility profile: enduring non-normative referents.

This profile is not independently required by any single requirement, but is structurally presupposed by the simultaneous satisfaction of R2–R5 under R1: stable reference to obligation-bearers, normative structures, occurrences, and applicability contexts requires referential anchoring to enduring entities that are neither normatively active nor temporally individuated.

Its necessity does not depend on denying that patients of actions could in principle bear obligations. Rather, under the admissibility criterion, collapsing acted-upon referents with obligation-bearing entities either makes normative attribution structurally admissible for all enduring referents or reintroduces the distinction through a discriminator at the extension layer. Either outcome is inadmissible under the neutrality and stability constraints adopted here.

4.2.7 Resulting Candidate Partition

The induced profiles are pairwise non-equivalent under R1's stability constraint. They differ either in persistence conditions or in admissible-relation classes.

Accordingly, any substrate satisfying R1–R6 must realize at least six distinct identity-and-persistence regimes. For reference, we fix the following abbreviations for these regimes:

1. **OBL: Obligation-bearing entity:** an enduring referent that may bear obligations or responsibilities and remain identifiable as a party across time.
2. **NOR: Normative structure:** an enduring referent that grounds obligations or permissions without being tied to any single occurrence.
3. **OCC: Time-indexed occurrence:** an entity individuated by its temporal realization and provenance.
4. **CTX: Applicability context:** a referent that scopes where, to whom, or under what conditions authority applies, and that may be nested or overlapping.
5. **REC: Descriptive record:** a referent that reports measured or derived properties without asserting causal or normative conclusions.
6. **ENR: Enduring non-normative referent:** an enduring entity that may be regulated, inspected, or otherwise affected by occurrences, but does not itself ground obligations or bear normative responsibility.

4.3 Collapse Admissibility Criterion

To establish a lower bound, we consider whether any two of the induced identity-and-persistence regimes can be collapsed into a single regime without violating the constraints of Section 2.

A collapse of two regimes is admissible only if the merged representation realizes a single extension-stable identity and persistence profile and supports a unified class of admissible relations without introducing interpretive flags, role predicates, or contextual discriminators at the extension layer, which would reintroduce regime distinctions contrary to the neutrality and extension-stability constraints of Section 2.

Under R1, identity and admissibility must be invariant across incompatible extensions. Accordingly, a collapse is inadmissible if it requires any of the following:

- **Identity instability:** the merged regime requires incompatible persistence criteria, so that identity judgments vary across extensions.
- **Category error:** the merged regime structurally admits relations inappropriate to some of its instances, requiring interpretive blocking rules.
- **Hidden regime reintroduction:** the original distinction is recovered through flags, roles, or contextual predicates, thereby reintroducing the regime at the extension layer.

Any collapse exhibiting one of these failure modes violates either extension-invariant identity or the neutrality criterion excluding hidden discriminators.

4.4 Pairwise Collapse Analysis

There are finitely many unordered pairs among the induced regimes. We consider each class of collapse and show that every such identification violates the admissibility criterion through one of the three failure modes defined above.

4.4.1 Category error: Obligation-bearing and Enduring non-normative referent

Suppose an obligation-bearing entity and an enduring non-normative referent are collapsed into a single regime.

The merged regime must admit a unified class of admissible relations. Obligation-bearing entities admit normative attribution relations such as *hasObligation* or *isResponsibleFor* without reclassification.

If those relations are structurally admissible for all instances of the merged regime, then every enduring referent becomes eligible for normative attribution in principle, embedding a normative commitment at the substrate level.

If, instead, normative attribution is restricted to only some instances, implementations must introduce a discriminator such as *isObligationBearer* to block inappropriate attributions. The original distinction is thereby reintroduced at the extension layer.

In either case, the collapse is inadmissible.

4.4.2 Identity instability: Obligation-bearing and Normative structure

Suppose an obligation-bearing entity and a normative structure are collapsed into a single regime.

Obligation-bearing entities persist across compliance, violation, and delegation. Normative structures persist across enactment, amendment, and repeal.

These persistence criteria are non-equivalent. No single identity condition can preserve both without making persistence judgments vary across extensions.

The collapse therefore violates extension-invariant identity under R1.

4.4.3 Category error: Normative structure and Applicability context

Suppose a normative structure and an applicability context are collapsed into a single regime.

Normative structures admit structural modification relations such as enactment, amendment, delegation, and repeal. Applicability contexts instead admit scoping relations such as nesting, overlap, and conditional refinement.

If the two are identified, the merged regime must admit both modification-type and scoping-type relations as structurally admissible. However, these relation classes serve fundamentally different roles. Amendment is a structural modification that preserves the identity of a normative structure across change. Whereas nesting and overlap are relations among scopes that organize applicability conditions without modifying normative content.

No single admissible-relation profile can unify these relation classes. Either the relations are admitted uniformly, producing category error for some instances, or their admissibility must be restricted contextually, reintroducing the distinction through a hidden discriminator.

If all relations are admitted uniformly, entities that are merely applicability scopes become amendable normative structures, producing category error. If relation admissibility is restricted by a predicate such as *isNormative* or *isScope*, the original distinction is reintroduced as a hidden regime at the extension layer.

If relation admissibility is restricted by a predicate such as *isNormative* or *isScope*, the original distinction is reintroduced as a hidden regime at the extension layer.

The collapse is therefore inadmissible.

4.4.4 Hidden-regime reintroduction: Normative structure and Descriptive record

Suppose a normative structure and a descriptive record are collapsed into a single regime.

Normative structures ground obligations and admit normative evaluation, enforcement, and modification. Descriptive records report measured or derived properties without asserting causal or normative commitments.

If both are identified, the merged regime must admit a unified admissible-relation profile. If normative evaluation relations are structurally admissible for all instances of the merged regime, then purely descriptive records become eligible for normative grounding, embedding evaluative commitments at the substrate level.

If, instead, normative evaluation is restricted to a subset of instances, implementations must introduce a discriminator such as *isNormative* or *isDescriptive* to distinguish grounding entities from descriptive ones. Such a discriminator reintroduces the original distinction at the extension layer, violating the admissibility criterion and the neutrality constraint.

The collapse is therefore inadmissible.

4.4.5 Summary of representative collapse failures.

Table 1 summarizes representative pairwise collapses and the failure mode each induces under the admissibility criterion defined in Section 4.3.

The remaining pairwise collapses instantiate the same three failure modes: identity instability, category error, or hidden-regime reintroduction. Under the admissibility criterion of Section 4.3, no collapse eliminates a regime without violating neutrality or stable reference (see Appendix 8 for the complete enumeration of all fifteen pairs).

Table 1: Representative collapse failures under neutrality and stability constraints

Collapsed regimes	**Failure mode**	**Reason for inadmissibility**
OBL / ENR	Category error or hidden-regime reintroduction	Normative attribution either becomes universally admissible (embedding a normative commitment) or must be blocked by a discriminator such as *isObligationBearer*, reintroducing the distinction at the extension layer.
OBL / NOR	Identity instability	Persistence as a responsible party and persistence as a normative structure require incompatible identity criteria across amendment and restructuring.
NOR / CTX	Category error	Modification relations (enactment, amendment, repeal) conflict structurally with scoping relations (nesting, overlap, conditional refinement); blocking one set requires a hidden discriminator.
NOR / REC	Hidden-regime reintroduction	Normative grounding and purely descriptive reporting admit incompatible admissibility profiles; implementations reintroduce *isNormative* or *isDescriptive* flags to recover the distinction.
OCC / ENR	Identity instability	Time-indexed individuation criteria conflict with extension-invariant persistence required for enduring entities.

The following section establishes that this bound is tight: a construction realizing exactly six regimes satisfies R1–R6 without hidden discriminators or additional identity categories, confirming that six is the minimum achievable structure under the stated constraints.

4.5 Lower-Bound Result

Lemma 4.1 (Lower bound on identity-and-persistence regimes)**.** *Under the assumptions of Section 2, any ontology satisfying requirements R1–R6 and the collapse admissibility criterion of Section 4.3 must realize at least six distinct identity-and-persistence regimes.*

Proof sketch. Section 4.2 shows that satisfying R1–R6 under extension-invariant identity induces six pairwise non-equivalent persistence-and-admissibility profiles.

Any ontology realizing fewer than six regimes must identify at least one unordered pair among these profiles.

By the collapse admissibility criterion (Section 4.3), every such identification produces either identity instability, category error, or hidden-regime reintroduction.

Each of these outcomes violates the neutrality or stability constraints fixed in Section 2.

Therefore no ontology with fewer than six regimes can satisfy R1–R6 simultaneously. □

4.6 Interpretation and Scope of the Necessity Result

The necessity result is conditional and structural. It does not claim that fewer regimes are impossible in general, but that they are incompatible with the optimization objective fixed in Section 2. Ontologies optimized for

local agreement, negotiated semantics, or expressive flexibility may legitimately employ fewer explicit distinctions by relying on hidden or interpretive mechanisms.

Within the stated constraints, however, the result establishes a minimal structural lower bound: any neutral substrate capable of supporting accountability under persistent disagreement must realize at least six distinct identity-and-persistence regimes. The following section shows that this bound is tight by exhibiting a construction that satisfies all requirements with a minimal set of six regimes.

5 Sufficiency: A Six-Regime Construction

5.1 The Six Identity-and-Persistence Regimes

We realize exactly the six identity-and-persistence regimes derived in Section 4. No additional regimes are introduced, and none of the induced regimes are merged or weakened.

Each regime is implemented with the identity conditions and admissible-relation profile specified in the derivation. The construction fixes these regimes as pairwise disjoint, with extension-invariant persistence criteria, and without interpretive flags or role-based discriminators.

We now specify a minimal set of typed relations that connects these regimes while preserving neutrality and stable reference.

5.2 Admissible Relations

A minimal set of typed, directional relations suffices to connect the six regimes and to satisfy all stated requirements. Representative relations include:

- authority grounding and delegation:

$$\textit{enacts},\ \textit{issues} \quad : \quad \mathrm{OBL} \rightarrow \mathrm{NOR}$$

- participation and party status:

$$\begin{array}{lcl} \textit{party-to} & : & \mathrm{OBL} \rightarrow \mathrm{NOR} \\ \textit{involves} & : & \mathrm{OCC} \rightarrow \mathrm{OBL} \\ \textit{has-patient} & : & \mathrm{OCC} \rightarrow \mathrm{ENR} \end{array}$$

- execution under authority:

$$\textit{occurs-under} \quad : \quad \mathrm{OCC} \rightarrow \mathrm{NOR}$$

- applicability:

$$\textit{applies-in} \quad : \quad \mathrm{NOR} \rightarrow \mathrm{CTX}$$

- description:

$$\textit{measures} \quad : \quad \mathrm{REC} \rightarrow (\mathrm{OBL} \mid \mathrm{ENR} \mid \mathrm{CTX})$$

Relations in this construction are structural and descriptive. They assert reference, participation, scoping, or provenance, but do not encode causal mechanism or normative validity.

Authority grounding and delegation record that an obligation-bearing entity enacts or issues a normative structure, without asserting validity or enforceability.

Participation relations record that an occurrence is associated with enduring referents in recorded participation relations. The relations *involves* and *has-patient* denote recorded association only; they do not assert agency, causal transformation, or normative responsibility.

The *measures* relation associates a descriptive record with an enduring referent, without asserting accuracy, causal relevance, or evaluative significance.

Occurrences serve as temporal and provenance anchors for participation and measurement relations. Descriptive records target enduring referents whose

identity persists across incompatible extensions.

This relation set suffices to satisfy R1–R6 without introducing additional identity regimes, role-based typing, or extension-layer discriminators.

Temporal and provenance information attach only to OCC and REC via typed attributes or admissible relations. These annotations do not introduce new identity criteria or additional identity-bearing categories.

5.3 Satisfaction of Structural Requirements

We briefly verify that the construction satisfies the requirements of Section 3:

- **R1 (Stable identity):** Each regime fixes an extension-invariant identity-and-persistence criterion.
- **R2 (Obligation-bearing capacity):** OBL admits normative attribution without reclassification.
- **R3 (Normative reference without execution):** NOR grounds authority independently of OCC.
- **R4 (Time-indexed occurrence):** OCC captures temporally individuated events.
- **R5 (Applicability and scope):** CTX provides first-class scoping referents supporting nesting and overlap.
- **R6 (Descriptive indicators):** REC represents measurements as stable referents distinct from OCC and NOR.

No additional identity regimes or extension-layer discriminators are required.

5.4 Sufficiency Proposition

Proposition 5.1 (Sufficiency of six regimes)**.** *The six-regime construction specified in this section satisfies R1–R6 under the assumptions of Section 2.*

Proof. By construction, each of R2–R6 is realized by a distinct identity-and-persistence regime, and all regimes satisfy R1. The regimes are pairwise disjoint, and their admissible relations introduce no cross-regime persistence dependencies. Hence satisfying the requirements jointly does not generate structural conflict. No additional regimes are required, and no interpretive flags or extension-layer discriminators are introduced. Non-redundancy of the construction follows because each of the six regimes realizes a persistence-and-admissibility profile shown in Lemma 4.1 to be non-collapsible: any identification of two regimes in this construction would instantiate one of the three failure modes established in Section 4.3, violating either extension-invariant identity or the neutrality constraint. □

5.5 Tightness of the Bound

Combining Lemma 4.1 and Proposition 5.1, establishes that exactly six identity-and-persistence regimes are necessary and sufficient for a neutral accountability substrate under the stated assumptions. The lower bound is therefore tight with respect to the optimization objective of interoperability under persistent disagreement. Any substrate realizing fewer regimes must either embed interpretive commitments at the substrate level, violating neutrality, or delegate essential identity-and-persistence distinctions to extension layers as hidden regimes, violating the admissibility criterion. Neither relaxation is available under the stated constraints, confirming that six is the minimum achievable structure, not merely a lower bound on an open range.

6 Interpretation of the Result

This section clarifies how the lower- and upper-bound results should be understood, and how they should not be understood. The aim is not to defend the result against alternative modeling goals, but to situate it relative to the explicit optimization criterion fixed earlier in the paper.

6.1 What the Result Establishes

Taken together, the necessity and sufficiency results establish a conditional optimality claim. Under the assumptions of Section 2, any ontology intended to function as a neutral substrate for accountability under persistent disagreement must realize at least six distinct identity-and-persistence regimes. Conversely, a construction with a minimal set of six such regimes suffices to meet all stated representational requirements without introducing hidden structure or interpretive commitments.

The result is therefore tight. The result does not merely show adequacy or inadequacy under particular conditions; it establishes that six regimes are both necessary and sufficient under the stated optimization objective.

Theorem 6.1 (Tightness of the six-regime bound)**.** *Under the assumptions of Sections 2–3, exactly six identity-and-persistence regimes are necessary and sufficient for a neutral accountability substrate under persistent disagreement.*

Proof. The theorem follows directly from Lemma 4.1 and Proposition 5.1. □

6.2 Minimality Relative to Stability

The result should not be read as a claim about simplicity in an intuitive or engineering sense. Ontologies with fewer explicit entity kinds may be easier to describe or extend within a coordinated interpretive community. However, when such reductions are achieved by encoding stability-critical distinctions as roles, flags, or contextual predicates, total structural complexity is not reduced. Instead, it is displaced into extension layers whose interpretation cannot be held fixed under disagreement.

In this paper, minimality is defined relative to stability. *A representation is minimal only if removing explicit structure does not require reintroducing the same distinctions implicitly.* From this perspective, an ontology with fewer explicit kinds but greater interpretive burden is not simpler, but less explicit.

6.3 Why Hidden Regimes Are Excluded

Hidden regimes are often effective where interpretation is shared, enforced, or low-cost to renegotiate. They support flexibility and expressive economy within a single governance framework.

The present analysis excludes such regimes not because they are incorrect, but because they are incompatible with the stated objective. When disagreement is persistent and uncoordinated, distinctions that are not fixed at the substrate level cannot be relied upon to preserve identity or admissible relations. Stability then depends on interpretive convergence, which the substrate is explicitly designed not to assume.

6.4 Relation to Other Ontological Strategies

Many upper ontologies and modeling frameworks adopt different optimization objectives, including metaphysical coverage, expressive richness, or alignment with natural language or scientific theory. Such frameworks may introduce additional kinds or tolerate contextual typing precisely in order to support richer modeling.

The result presented here neither subsumes nor contradicts those approaches. It identifies a specific point in the design space: the minimal structure required for a neutral, extension-stable substrate under persistent disagreement. Frameworks optimized for other goals may legitimately occupy different points in that space.

Table 2 illustrates common representation strategies for these regimes in BFO 2.0 and DOLCE and highlights where regime-level distinctions required for neutrality are not structurally enforced.

The comparison is structural rather than taxonomic; it identifies where regime-level distinctions required for neutrality are left to implementation rather than enforced at the type level.

The table identifies the structural under-constraint that would arise in each case. These under-constraints are not problems in their intended use. They are points where neutral substrate implementations built on those frameworks might need to introduce additional modeling commitments that the present substrate treats as regime-level structure.

The six regimes are not derived from gaps in existing frameworks. Rather, independently motivated requirements reveal where those frameworks leave regime-level distinctions to the implementer.

Table 3 summarizes the systematic under-constraints that arise in both frameworks relative to the neutrality and stability requirements.

6.5 Conditional Scope of the Claim

The lower-bound result is explicitly conditional. If any assumption of Section 2 is rejected, such as neutrality, stability under incompatible extension, or the exclusion of hidden regimes, the optimization criterion is invalidated and the bound no longer applies. We do not claim that these assumptions are universally correct; we claim only that the result follows when they are adopted.

If stability is weakened to allow context-dependent regime membership (e.g., role- or viewpoint-relative identity/admissibility), some regimes may be representable as role- or parameter-dependent refinements of fewer base regimes, but doing so shifts structural burden into coordination, versioning, or exchange mechanisms whose admissibility conditions are not fixed at the substrate level and are outside the scope of this analysis; the lower bound therefore does not apply under such weakenings.

Within the stated scope, however, the result is unavoidable. Under the stated assumptions, any attempt to reduce the number of regimes would require sacrificing stability, reintroducing hidden distinctions, or weakening the neutrality constraint. The six-regime structure thus emerges not as a

Table 2: Illustrative representation strategies for the six regimes in BFO 2.0 and DOLCE, with structural under-constraints under neutrality and stability requirements.

Regime	**BFO 2.0**	**DOLCE**
ENR (enduring non-normative referent)	Independent Continuant (material object). Clean for material entities; institutional and fiat entities require additional modeling choices not enforced by the framework.	Physical or Social Object (endurant). Social object is first-class, providing a cleaner fit for institutional referents than BFO.
OBL (obligation-bearing entity)	Specifically Dependent Continuant (Role or Disposition) inhering in an Independent Continuant. Obligation-bearing is not a first-class category; it is realized via a role–bearer pair, splitting what the present substrate treats as a single regime.	Quality or property of a Social Object. Obligation-bearing capacity is not a structural first-class distinction; the OBL/NOR boundary remains implementation-dependent.
NOR (normative structure)	Generically Dependent Continuant (information artifact, normative document). The substrate-level separation between authority-as-referent (NOR) and authority-as-executed (OCC) is a modeling choice, not structurally enforced.	Social Object (normative structure). Better motivated than BFO, but authority and execution can both be represented as social objects or perdurants without a regime-level firewall between NOR and OCC.
OCC (time-indexed occurrence)	Process or Process Boundary (occurrent). Closest mapping in both frameworks; BFO occurrents are well-individuated temporally.	Perdurant (Event or Process). DOLCE distinguishes achievements from activities, which is richer than required but not incompatible.
CTX (applicability context)	Site (immaterial Independent Continuant) for spatial scope only. Non-spatial applicability contexts (normative jurisdiction, conditional scope) have no stable BFO home and collapse into GDC, conflating CTX with NOR.	Abstract Region or Feature. DOLCE regions are spatiotemporal; normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
REC (descriptive indicators without causal commitment)	Generically Dependent Continuant (data item, measurement information entity). Both normative structures (NOR) and descriptive indicators (REC) map to GDC; the stability-critical distinction between them is invisible at the BFO type level.	Quality (quale or abstract region). DOLCE's quality theory is the strongest match; the without-causal-commitment constraint is a modeling discipline, however, not structurally enforced.

design preference, but as a structural consequence of the problem as posed.

The present analysis does not model state transitions, versioning mechanisms, or exchange protocols. Such structures may be required for practical interoperability, but they operate downstream of the regime-level identity constraints established here.

6.6 Implications

The primary implication of this result is diagnostic rather than prescriptive. It provides a principled way to evaluate claims that an ontology supports neutral interoperability under disagreement. If such a framework relies on implicit

Table 3: Systematic under-constraints in BFO 2.0 and DOLCE relative to neutrality and stability requirements.

Under-constraint	**BFO 2.0**	**DOLCE**
NOR/REC conflated	Both normative structures and descriptive indicators map to GDC; the stability-critical distinction between them is invisible at the type level and left to the implementer.	Both map to Social Object or Quality depending on implementation; the distinction is a modeling discipline, not structurally enforced.
CTX has no structural home	Non-spatial applicability contexts collapse into GDC, conflating CTX with NOR.	Normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
OBL/NOR boundary implementation-dependent	The separation between obligation-bearer and normative structure is mediated by the inheres-in relation, not enforced by regime membership; unstable under delegated authority.	The same boundary is mediated by participation relations; regime-level enforcement is absent.

Implementations that resolve these under-constraints silently introduce hidden regimes in exactly the sense of Section 4.

roles or context-dependent typing for stability-critical distinctions, the present result explains why those claims cannot be sustained simultaneously under the stated constraints.

Conversely, for designers who share the stated optimization objective, the result identifies which distinctions must be made explicit, and which may safely be deferred to extension layers, in order to preserve neutrality and stable reference across incompatible interpretations. The six-regime structure thus serves both as a lower bound and as a design template for accountability substrates in contested domains.

The resulting regime structure can be understood as an axiomatization pattern for substrates intended to support divergent specialization: identity and admissibility constraints are fixed at the substrate level so that subsequent causal or normative extensions may diverge without retroactive reinterpretation.

6.7 Grounding the Construction in the Running Cases

The six-regime construction is not an abstract partition. Each case introduced in Section 1 exercises the full structure and illustrates what stable reference under persistent disagreement concretely requires.

6.7.1 Multi-jurisdiction regulatory reporting.

A corporate entity is represented as an obligation-bearing referent (OBL) that persists across jurisdictional differences in how corporate personhood or subsidiary liability is defined. The reporting statute under which it operates is represented as an authority-bearing structure (NOR) that persists across amendments, contested interpretations, and delayed enforcement. Each filing, inspection, or transaction is represented as a time-indexed occurrence (OCC) individuated by when and where it happened, independently of how any jurisdiction evaluates it. The jurisdiction itself is represented as an

applicability context (CTX) that scopes which obligations apply to which entities without collapsing into either the statute or the filer. Performance and compliance indicators are represented as descriptive records (REC) that remain comparable across jurisdictions without carrying evaluative or causal commitments. Any entity that is regulated but does not itself bear obligations, such as a regulated substance or a monitored facility, is represented as an acted-upon referent (ENR). The six-regime construction allows two jurisdictions to refer to the same corporation, the same statute, and the same filing while maintaining incompatible evaluations of compliance, without either forcing interpretive convergence or losing the shared referential ground required for audit comparison.

6.7.2 Legislative history tracking.

A statute is represented as an authority-bearing structure (NOR) that remains the same normative referent across the legislative acts that enacted it (OCC), the amendments that modified it (OCC), and the administrative decisions that applied it (OCC). The legislature and the agencies that implement the statute are represented as obligation-bearing referents (OBL) whose identity does not depend on any particular enactment event. Jurisdictions and conditional scopes such as the geographic or subject-matter boundaries of the statute's application, are represented as applicability contexts (CTX) that may be nested, overlapping, or contested without destabilizing reference to the statute itself. Records of legislative proceedings, judicial interpretations, and compliance histories are represented as descriptive indicators (REC) that characterize outcomes without asserting which interpretation of the statute is correct. The construction allows legal historians, policy analysts, and regulators working under incompatible interpretive frameworks to refer to the same statutory authority and the same sequence of enactment events while disagreeing entirely on what the statute means or which application was legitimate.

6.7.3 Cross-institutional provenance.

A research or administrative record is represented as an acted-upon referent (ENR) whose identity and persistence conditions are invariant under the classificatory schemes of any receiving institution. The institution that produced the record and the institutions that subsequently hold or analyze it are represented as obligation-bearing referents (OBL) that remain identifiable as custodians across transfers, mergers, and re-organizations. Policies governing data use, retention, or sharing are represented as authority-bearing structures (NOR) that persist independently of whether they have been enforced or contested in any particular transfer. Each transfer, annotation event, or re-analysis is represented as a time-indexed occurrence (OCC) individuated by its provenance, independently of how institutions classify the action. Institutional scope conditions, such as which data a policy governs or which entities fall under a data-sharing agreement, are represented as applicability contexts (CTX). Derived indicators, quality metrics, and outcome summaries attached to the record are represented as descriptive records (REC) that remain comparable across institutions whose causal and evaluative frameworks differ. The construction preserves the provenance chain as a stable referential structure even when no two institutions agree on how the record should be classified, evaluated, or interpreted.

7 Related Work

This paper is situated at the intersection of formal ontology, knowledge representation, and the study of classification under disagreement. Rather than surveying ontology engineering broadly, it situates the present result with respect to prior work on neutrality, identity, and structural stability.

7.1 Classification, Disagreement, and Neutrality

A substantial body of work has shown that classification systems embed institutional, social, and political commitments rather than functioning as neutral descriptive devices (Bowker and Star 1999; Haslanger 2012). In scientific and policy contexts, disagreement is often persistent rather than transient, reflecting incompatible evaluative frameworks rather than resolvable empirical uncertainty (Longino 1990).

These observations motivate the need for representations that remain usable without assuming interpretive convergence. The present work adopts this motivation but addresses a distinct question: what minimal structural commitments are unavoidable when disagreement is treated as permanent and interpretive commitments are externalized rather than negotiated?

7.2 Identity, Roles, and Ontological Methodology

Formal ontology has long emphasized the role of identity conditions and persistence criteria in category formation. The OntoClean methodology (Guarino and Welty 2002) clarifies why roles are anti-rigid and unsuitable as identity-defining categories (Guarino 1998), and subsequent work on social and institutional entities has further developed these distinctions (Masolo et al. 2003).

This paper builds on these insights but applies them at the substrate level. The present analysis operates at a different level than role-based modeling, deriving substrate-level identity-and-persistence regimes from neutrality requirements rather than from domain classifications.

7.3 Upper Ontologies

Upper ontologies such as BFO (Smith and Ceusters 2015), DOLCE (Gangemi et al. 2002), and UFO (Guizzardi 2005) provide rich accounts of dependence,

endurants and perdurants, and social objects. They differ substantially in commitments and optimization objectives, and are not designed primarily as neutral substrates for interoperability under persistent disagreement.

The present work does not compare expressivity or advocate alignment with any particular upper ontology. Instead, it derives lower-bound constraints that apply independently of such frameworks, conditional on the neutrality and stability requirements adopted here. The regime structure derived here can be situated with respect to two widely-used frameworks to clarify where under-constraints arise not as defects in those frameworks but as points where implementations built on them would need to introduce additional commitments to satisfy the neutrality and stability requirements assumed here.

BFO (Smith and Ceusters 2015) organizes entities under the continuant/occurrent distinction: continuants persist through time while occurrents unfold across it (Arp et al. 2015b). OCC (time-indexed occurrences) maps cleanly onto BFO occurrents. ENR through NOR map onto independent and generically dependent continuants, but BFO does not enforce the regime-level separation between them: obligation-bearing entities (ENR), authority-bearing structures (NOR), and descriptive indicators (REC) are all candidates for Generically Dependent Continuant, leaving their stability-critical distinctions to the implementer. CTX (applicability and scope) has no first-class BFO home; non-spatial applicability contexts such as normative jurisdiction or conditional scope collapse into GDC, conflating CTX with NOR under a BFO-based implementation. Under persistent disagreement, these conflations require hidden discriminators to recover regime-level distinctions, introducing implicit structure of precisely the kind the neutrality constraint prohibits.

DOLCE (Gangemi et al. 2002) introduces social objects as first-class entities and provides a richer treatment of institutional and normative structure than BFO (Masolo et al. 2003). This makes DOLCE a better fit for OBL and NOR as distinct categories. However, the NOR/REC conflation persists: both normative structures and descriptive indicators are candidates for so-

cial object or quality, and the distinction between authority-as-referent and descriptive-record-as-referent is not enforced at the type level. DOLCE's region-based account of scope is spatiotemporal, leaving normative applicability contexts (CTX) without a structural home distinct from NOR or REC, and the OBL/NOR boundary (between obligation-bearer and normative structure) is mediated by participation relations rather than enforced by regime membership (Masolo et al. 2003). As with BFO, implementations resolving these under-constraints under persistent disagreement silently introduce hidden regimes.

These observations do not constitute objections to BFO or DOLCE as general-purpose ontological frameworks. They locate a specific point where the optimization objectives of those frameworks (expressiveness, metaphysical coverage, alignment with scientific practice) diverge from the optimization objective assumed here, and where that divergence has structural consequences for neutral interoperability under persistent disagreement.

When constrained to enforce conservative extension over identity and admissibility conditions, such frameworks can realize the regime structure derived here. The contribution of this paper is to identify the structural constraints required when neutrality under persistent disagreement is taken as the primary objective.

The present result is therefore not a competing upper ontology, but a constraint on any ontology intended to function as a neutral, stability-preserving substrate under persistent disagreement.

7.4 Relation to Common Logic and ontology exchange

The neutral substrate and derived regimes can be expressed in ISO/IEC 24707 Common Logic (ISO/IEC 2018) as a conservative theory, supporting interpreter-independent exchange without reliance on description-logic restrictions or fixed signatures. Here, conservative means that extensions add structure without altering the consequences derivable about substrate-level

terms (Hodges 1997), ensuring that incompatible interpretive frameworks do not retroactively change the meaning of shared references. This enables compatibility with ontology exchange workflows widely used in applied ontology (Kutz et al. 2010) and digital twin infrastructures (Grieves and Vickers 2016), where multiple interpretive layers must interoperate over stable referents.

7.5 Causality, Explanation, and Infrastructure

Separating descriptive structure from causal or explanatory claims is a well-established principle in causal analysis (Pearl 2009). Similarly, research on large-scale information infrastructures emphasizes the importance of long-term stability and interpretive restraint across institutional boundaries (Edwards 2011).

These perspectives align with the pre-causal and pre-normative constraints assumed in this paper. The ontology analyzed here is not an explanatory theory, but an infrastructural substrate intended to support explanation, evaluation, and policy analysis without embedding them.

7.5.1 Relation to FAIR Data Principles

The FAIR principles promote data reuse across communities and over time, but do not specify the ontological constraints required for interoperability under persistent disagreement. In practice, reuse is often pursued through negotiated vocabularies or shared interpretations, which presuppose forms of convergence that may not be available in legal, political, or analytic settings.

The results of this paper clarify a structural precondition for FAIR-aligned reuse in pluralistic or institutionally heterogeneous contexts: stable reference must be supported independently of causal, normative, or evaluative agreement. The identity-and-persistence regimes derived here therefore constrain the ontological substrates upon which FAIR-compliant systems implicitly

rely when reuse occurs under persistent disagreement.

8 Conclusion

This paper has addressed a structural question that arises when ontologies are required to function as neutral substrates under persistent disagreement. Under explicit constraints of neutrality, stability under incompatible extension, and the exclusion of hidden or interpretive distinctions, we have shown that a lower bound exists on the ontological structure required to support accountability. Ontologies with fewer than six distinct identity-and-persistence regimes cannot satisfy the stated neutrality and stability requirements under the assumptions fixed in Section 2; a construction with exactly six regimes suffices without excess structure.

The result is conditional rather than prescriptive. It does not claim that the identified structure is optimal for all applications, nor does it assess the adequacy of alternative modeling strategies for other objectives. It establishes only that, under a particular and explicitly stated optimization objective, certain distinctions are structurally unavoidable.

This work clarifies the design space for neutral substrates under persistent disagreement. Under the stated constraints, it shows that minimality is fixed by structural requirements rather than by modeling preference. Future work may examine how alternative optimization objectives yield different structural bounds, or analyze how the six-regime construction behaves when stability or neutrality assumptions are systematically weakened.

Acknowledgements

The author thanks the anonymous reviewers for detailed and technical engagement with the argument. Their challenges substantially strengthened the necessity proof and clarified the scope of the central result.

Portions of this work were developed using computer-assisted tools during manuscript preparation. Generative language models were used to assist with editing, formatting, and consistency checking during manuscript preparation. All conceptual framing, formal development, results, interpretations, and conclusions are the author's own. All generated suggestions were critically reviewed and validated, and the author takes full responsibility for the content of this work.

Statements and Declarations

Author Contributions

The author was solely responsible for the conception, analysis, and writing of this manuscript.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

This research received no external funding.

Data Availability

No datasets were generated or analyzed during the current study.

Ethical Approval

Not applicable.

Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

Appendix A. Complete Pairwise Collapse Enumeration

All $\binom{6}{2} = 15$ unordered pairs among the six induced regimes are classified below by failure mode under the admissibility criterion of Section 4.3.

Identity Instability

Five pairs cannot be collapsed without violating extension-invariant identity:

- OBL/NOR (incompatible persistence across delegation and amendment);
- OBL/OCC (obligation-bearing continuity conflicts with occurrence closure at temporal realization);
- OCC/ENR (time-indexed individuation conflicts with enduring persistence);
- OCC/NOR (temporal realization conflicts with normative persistence across repeal);
- OCC/CTX (time-indexed individuation conflicts with scope persistence).

Category Error

Six pairs cannot be collapsed without producing a category error under the neutrality constraint:

- OBL/ENR (normative attribution becomes universally admissible or requires blocking discriminator);
- OBL/CTX (obligation-bearing and scoping admit incompatible relation profiles);
- NOR/CTX (modification and scoping relations admit no unified pro-

file);

- NOR/ENR (normative grounding relations misclassify purely enduring referents);
- CTX/ENR (scoping relations conflict with unscoped enduring persistence);
- CTX/REC (scoping relations misclassify descriptive records as applicability contexts).

Hidden-regime Reintroduction

Four pairs cannot be collapsed without reintroducing the original distinction as a hidden regime:

- NOR/REC (normative grounding requires *isNormative* discriminator to protect descriptive records);
- OBL/REC (normative attribution requires discriminator to exclude descriptive instances);
- ENR/REC (measurement targets and descriptive records require *isDescriptive* flag to distinguish them);
- OCC/REC (provenance discriminator required to separate occurrence anchors from reports).

Appendix B. Glossary of Terms

This appendix defines key terms used throughout the paper.

Glossary

Accountability Substrate

A representational layer intended to support attribution, inspection, and comparison of responsibility-related structures without embedding causal, normative, or evaluative commitments. In this paper, accountability is treated structurally rather than interpretively.

Applicability Context

An enduring referent that delimits where, to whom, or under what conditions an authority-bearing structure applies. Applicability contexts persist independently of specific occurrences and must support nesting and overlap.

Hidden Regime

An identity- or persistence-relevant distinction that is not represented as an explicit ontological kind, but is instead encoded via roles, flags, contextual predicates, or interpretive conventions supplied by extension layers. Hidden regimes are excluded under the stability optimization adopted here.

Identity-and-Persistence Regime

A specification of the conditions under which an entity remains the same entity over time and across incompatible extensions, together with the relations it may admit without reclassification. Each regime defines a distinct mode of reference stability.

Interpretive Extension

Any explanatory, normative, analytic, or evaluative structure added

atop the substrate that supplies meaning, causation, or judgment without altering substrate identity conditions.

Lower-Bound Result

A necessity claim establishing that no representation with fewer than a specified number of identity-and-persistence regimes can satisfy the stated requirements under the adopted optimization criterion.

Neutrality Constraint

The requirement that the substrate itself remain pre-causal and pre-normative, permitting incompatible interpretations without requiring revision of entity identity or structure.

Optimization Criterion

The objective function fixed for this paper: to minimize explicit ontological structure subject to neutrality, stable reference, and the exclusion of hidden identity regimes. Minimality is defined relative to stability, not syntactic brevity.

Persistent Disagreement

A condition in which interpretive, normative, or analytic incompatibilities are expected to endure over time, rather than converge through negotiation, evidence, or governance.

Stable Reference

The property that entities remain referable as the same entities across time and across incompatible extensions, without reclassification or reinterpretation of identity criteria.

Sufficiency Construction

An explicit ontological partition that satisfies all stated requirements using exactly the minimal number of identity-and-persistence regimes established by the lower-bound result.

Tight Bound

A result showing that the same number of regimes is both neces-

sary and sufficient under the stated assumptions and optimization objective.

Time-Indexed Occurrence

An entity whose identity is inseparable from its temporal realization and provenance. Occurrences do not persist beyond their happening and are distinct from enduring entities.

References

Robert Arp, Barry Smith, and Andrew D. Spear. *Building Ontologies with Basic Formal Ontology*. MIT Press, 2015a.

Robert Arp, Barry Smith, and Andrew D Spear. *Building ontologies with basic formal ontology*. Mit Press, 2015b.

Geoffrey C. Bowker and Susan Leigh Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999.

Denise M. Case. The ontological neutrality theorem: Why neutral ontological substrates must be pre-causal and pre-normative, 2026. URL `https://arxiv.org/abs/2601.14271`.

Paul N. Edwards. *A Vast Machine: Computer Models, Climate Data, and the Politics of Global Warming*. MIT Press, 2011.

Aldo Gangemi, Nicola Guarino, Claudio Masolo, Alessandro Oltramari, and Luc Schneider. Sweetening ontologies with dolce. In *Proceedings of the 13th International Conference on Knowledge Engineering and Knowledge Management (EKAW 2002)*, volume 2473 of *Lecture Notes in Computer Science*, pages 166–181. Springer, 2002. doi: 10.1007/3-540-45810-7_18.

Michael Grieves and John Vickers. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary perspectives on complex systems: New findings and approaches*, pages 85–113. Springer, 2016.

Nicola Guarino. On the ontological status of roles. *Applied Ontology*, 1998.

Nicola Guarino and Christopher A Welty. Evaluating ontological decisions with OntoClean. *Communications of the ACM*, 45(2):61–65, 2002.

Giancarlo Guizzardi. *Ontological Foundations for Structural Conceptual Models*. PhD thesis, University of Twente, Enschede, The Netherlands, 2005.

Sally Haslanger. *Resisting Reality: Social Construction and Social Critique*. Oxford University Press, Oxford, 2012. ISBN 9780199577455.

Wilfrid Hodges. *A Shorter Model Theory*. Cambridge University Press, 1997.

ISO/IEC. Information technology — common logic (CL): A framework for a family of logic-based languages. Technical Report ISO/IEC 24707:2018, International Organization for Standardization, 2018.

Nele Köhler and Fabian Neuhaus. The mercurial top-level ontology of large language models. *Applied Ontology*, 20(1):5–15, 2025.

Oliver Kutz, Till Mossakowski, and Dominik Lücke. Carnap, goguen, and the hyperontologies: Logical pluralism and heterogeneous structuring in ontology design. *Logica Universalis*, 4(2):255–333, 2010.

Timothy Lebo, Satya Sahoo, and Deborah McGuinness. PROV-O: The PROV ontology. Technical report, W3C, 2013. URL `https://www.w3.org/TR/prov-o/`. W3C Recommendation.

Helen E. Longino. *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry*. Princeton University Press, Princeton, NJ, 1990. ISBN 9780691020512.

Claudio Masolo, Stefano Borgo, Aldo Gangemi, Nicola Guarino, and Alessandro Oltramari. Wonderweb deliverable d18: Ontology library. In *IST Project 2001-33052 WonderWeb: Ontology Infrastructure for the Semantic Web*, 2003.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

Barry Smith and Werner Ceusters. Basic formal ontology 2.0: Specification and user's guide. *Applied Ontology*, 10(1):1–77, 2015. doi: 10.3233/AO-150145.